

ASSIGNMENT – III

III . Inverse Laplace Transforms using Partial Fractions

1. Find $L^{-1} \left[\frac{2s^2 - 6s + 5}{s^3 - 6s^2 + 11s - 6} \right]$

Ans : $f(t) = \frac{1}{2}e^t - e^{2t} + \frac{5}{2}e^{3t}$

2. Find $L^{-1} \left[\frac{2s^2}{(s^2+1)(s-1)^2} \right]$

Ans : $f(t) = -\cos t + e^t + te^t$

3. Find $L^{-1} \left[\frac{5s+3}{(s-1)(s^2+2s+5)} \right]$

Ans : $f(t) = e^t + e^{-t} \left\{ \frac{3}{2} \sin 2t - \cos 2t \right\}$

4. Find $L^{-1} \left[\frac{s+2}{(s^2+4s+8)^2} \right]$

Ans : $f(t) = \frac{1}{2}te^{-2t} \sin 2t$

5. Find $L^{-1} \left[\frac{s}{s^4+4a^4} \right]$

Ans : $f(t) = \frac{\sin at \sinh at}{2a^2}$